Experiment - 11

Aim

To design and configure a simple internetwork using a router in cisco packet tracer.

Procedure:

1) Open cisco packet tracer and create a new topology.

2) Add 2 routers, switches and 3-4 PC's for each network.

3) Connect PC's to switches & routers using ethernet cables

4) Assign static IP address and subnet mask to all PC's

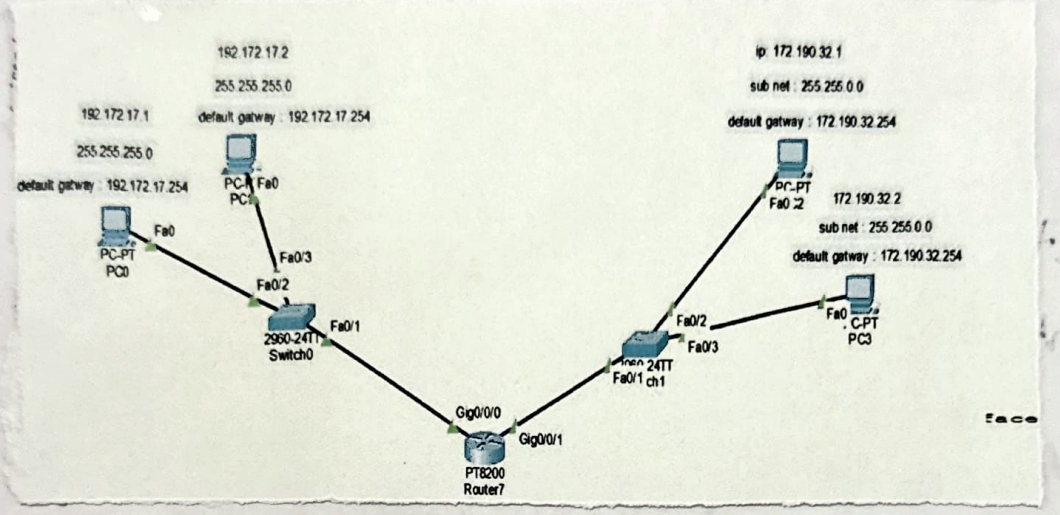
Code:

```
enable
configure terminal
interface Fast Ethernet 0/0
ip address <IP> <Subnet mask>
no shut down
exit
interface Fast Ethernet 0/1
no shutdown
exit.
```

6) Configure default gateway of each PC with IP address of connected router interfaces.

7) Use simulation Mode and send a PDU from one network to another

8) Observe the packet path through the router and verify the successful communication.




```
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/2.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```

Result..

The simple subnetwork was successfully designed and configured in Cisco packet tracer.